# DSA with C++

**Year: I** **Semester: I**

| Unit | Topics |
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| 1 | **Introduction to components of a computer system**: Memory, processor, I/O devices, storage, and operating system. Concept of assembler, compiler, interpreter, loader and linker.<br><br>**Idea of program/algorithm**: Representation of algorithms - flowchart, and algorithms. Visualization of flowchart and algorithms. From algorithms to programs, and source Code.<br><br>**Programming basics:** Writing and executing the first C++ program, syntax and logical errors in compilation with examples. Object and executable code. The structure of a C++ program, the main function, header files, standard output/input, and comments. |
| 2 | **Tokens in C++:** Keywords, identifiers/variables, constants, punctuators, and operators.<br><br>**Variables:** Variable naming convention, scope and their lifetime.<br><br>**Data types:** Data types classification - basic, derived and user defined. Basic data types: Data types, their storage requirements, and typecasting.<br><br>**Operators:** Arithmetic and miscellaneous operators, relational operators, logical operators, conditional operators, and bitwise operators. Associativity and precedence. Expressions and their evaluation.<br><br>Introduction and usage of Math library functions. |

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| | **Taking input on coding platforms - 1**: Single value in a single line, multiple known values in a single line, multiple known values in multiple lines.<br><br>**Program flow control**: if statement, if - else statement, nested if - else, else - if ladder, and switch - case. |
| 3 | **Iteration and loops**: Use of while, do-while and for loops, multiple loop variables, use of break, and continue statements. Nested iteration.<br><br>**Taking input on coding platforms - 2**: Single value in a single line, multiple known values in a single line, and others.<br><br>**Arrays:** Array notation and representation, manipulating array elements, using multi dimensional arrays. Introduction to dynamically growable arrays (vectors in STL),<br><br>**Taking inputs on coding platforms - 3**: Multiple unknown values in a single line, multiple unknown values in multiple lines. |
| 4 | **Functions**: Introduction, types of functions, passing parameters to functions, call by value, call by reference. Introduction to recursive functions. |
| 5 | **Strings**: String representation in C++. String manipulation operations with the help of functions from string.h library, specifically, replacing, inserting and erasing characters, finding character or substrings, counting characters or substrings, finding length of a string, case conversion, dealing with alphanumeric strings, concatenating two or more strings, and comparing two strings.<br><br>**Introduction to Object Oriented Programming**: Discussion on characteristics of OOP, basic understanding of class and objects with examples, properties and methods in a class. Basic structure of a class in C++ and creating objects. |